

MATHEUS HOSTERT

502 Van Allen Hall, Iowa City, IA 52242

matheus-hostert@uiowa.edu | mhostert.com | ORCID: 0000-0002-9584-8877 | INSPIRE: Matheus.Hostert.1

ACADEMIC POSITIONS

- University of Iowa** Aug. 2025 - present
Assistant Professor.
- Harvard University** Sept. 2023 - Aug. 2025
Neutrino Theory Network Fellow. Department of Energy Fellowship.
- University of Minnesota & Perimeter Institute** Sept. 2019 - Sept. 2023
Joint four-year postdoctoral researcher appointment.

EDUCATION

- Ph.D.** in Theoretical Physics – Durham University, United Kingdom Oct. 2015 - Sep. 2019
Dissertation: Hidden Physics at the Neutrino Frontier: Tridents, Dark Forces, and Hidden Particles.
Supervisor: Prof. Silvia Pascoli.
Dissertation Committee: Profs. Joachim Kopp and David Cerdeño.
- Bachelor's degree** in Physics – Federal University of Santa Catarina, Brazil Mar. 2011 - Jun. 2015
With a year abroad at Durham University (Sept. 2013 - Sept. 2014) and honors in advanced mathematics.

FELLOWSHIPS AND AWARDS

- Old Gold Fellowship** (Apr. 2026): developmental program for tenure-track assistant and associate professors at the University of Iowa.
- Neutrino Theory Network fellowship** (Jan. 2023): fully funded postdoctoral position at Harvard University.
- Science without Borders Ph.D. scholarship** (Sept. 2015): excellence-based Brazilian scholarship for a full Ph.D. abroad.
- Science without Borders Undergraduate scholarship** (Sept. 2013): excellence-based Brazilian scholarship for one year of undergraduate studies abroad.

TEACHING

- **Courses Taught:**
 - Graduate-level special topics course on Particle Physics Beyond-the-Standard Model, Fall 2025.
 - Graduate-level Quantum Mechanics II, Spring 2026.
- **Other teaching:**
 - **International Neutrino Summer School 2026** (*upcoming*, Summer 2026): lectures on “Beyond the Standard Model with Neutrinos”.
 - **Fermilab Neutrino University**: lecture on “New Physics with Neutrinos” as part of the Fermilab lecture series for summer interns.
 - **ICTP-SAIFR School on Dark Matter and Neutrino Detection, 2024**: three lectures on neutrinos beyond the Standard Model at ICTP-SAIFR in São Paulo.
 - **Perimeter ISSYP 2022 and 2023**: led high-school students in a one-week project about neutrinos.

SUPERVISION

- **Post-doctoral scholars:**
 - Dr. Stephan Meighen-Berger (Fall 2025 – present), neutrinos and dark matter.
 - Dr. Aparajitha Karthikeyan, Neutrino Theory Network Postdoctoral Fellow (starting Fall 2026), neutrinos and dark matter.
- **Ph.D. students:**
 - Joel Choi, working on accelerator neutrinos and muon colliders.
 - Esmenaz Isik, working on high-energy astrophysics and neutrino telescopes.
- **Undergraduate students:**
 - Yatharth Sirohi, Fall 2025, Spring 2026, and Charles A. Wert Summer Research Grant 2026.
 - Landen Freeman, Fall 2025, Spring 2026, and Charles A. Wert Summer Research Grant 2026.
 - Megan Murphy, Fall 2025, Spring 2026, and Independent Creative Research by Undergraduates (ICRU) Fellowship 2026.
- **Postdoc mentoring in Ph.D. projects:**
 - Jaime Hoefken (University of Bologna, 2022 to 2023): two publications.
 - Daniele Massaro (University of Bologna, 2021 to 2023): four publications.
 - Asli Abdullahi (Durham University, 2020 to 2021): two publications.
 - Nicholas Kamp (MIT, 2021 to 2022): two publications,
 - Nicolò Foppiani (Harvard University, 2020 to 2022): two publications.
- **Postdoc mentoring in undergraduate projects:**
 - Isaac Cheng (University of Waterloo, 2023): primary advisor in PSI Start Summer project at the Perimeter Institute working on the cosmic neutrino background asymmetry.
 - Nathan Gershengorn (Harvard University, 2024): research on the effect of matter potentials on the cosmic neutrino background asymmetry,
 - Luc Bojorquez-Lopez (Harvard University, 2024): research on the beam-induced neutrino rate at muon collider detectors, resulting in a publication.
 - Connor Sponsler (Harvard University, 2024): research on new matter resonances in neutrino propagation through the atmosphere, resulting in a publication.

SCIENCE OUTREACH

- **Media coverage:** Research and quotes featured in Quanta Magazine, The New York Times, and Wired.
- **IceCube Masterclass 2024:** main local organizer at Harvard for the IceCube Masterclass 2024.
- **Perimeter ISSYP 2022 and 2023:** tutor at two editions of the International Summer School for Young Physicists at Perimeter.
- **KITP Teacher's Conference 2022:** speaker at the KITP teacher's conference.
- **IceCube comics:** translated all the Rosie & Gibbs comics about IceCube into Brazilian Portuguese.

- **Royal Society Summer Exhibition 2017 and 2018:** event organizer for the “Modeling the Invisible” exhibition and the “Ghosts in the Universe” exhibition.
- **Celebrate Science 2018:** volunteer in regional outreach event for schools in County Durham.
- **Orkney Science Festival 2018:** volunteer in the International Orkney Science Festival, visiting schools in remote islands of the Orkney archipelago in the north of Scotland.
- **Pint of Science 2017:** event manager for a local outreach event in County Durham.

COMMUNITY ENGAGEMENT

- **Violation of Fundamental Symmetries with B mesons:** co-convener for the Dark Sector sessions.
 - **INSS 2025:** member of the local organizing committee for International Neutrino Summer School (INSS) 2025.
 - **NuFact Workshop:** convener for “Neutrinos Beyond PMNS” working group under a three-year term (2024 - 2026).
 - **Snowmass 2021:** Editor for the “Neutrino Frontier” white paper on sterile neutrinos and the “Rare Processes and Precision Measurements” white paper on new physics in kaon and hyperon factories. Made substantial contributions to over seven white papers and led a letter of intent on non-minimal neutrino mass models.
 - **CERN FPC PBC:** member of the Feebly Interacting Particle (FPC) working group, part of the Physics Beyond Colliders (PBC) effort at CERN.
 - **Weak Interactions and Neutrinos 2021:** one of primary event organizers for the 2021 edition in Minnesota, US.
- IceDUNE workshop 2021:** convener of the Beyond the Standard Model physics session aimed at increasing the cross-talk between IceCube and DUNE collaborators.
- **Young Theorists Forums 9, 10, and 11:** event organizer for the annual one-day conference for graduate students in the UK.
 - **Journal refereeing:** refereed for Physical Review Letters, Physical Review D, Journal of High Energy Physics, European Physics Journal C, and the Astrophysical Journal.

EXPERIMENTAL COLLABORATIONS

- Participated in a Memorandum of Understanding to collaborate with **MicroBooNE** on a search for dark sectors.
- Member of the **Deep Underground Neutrino Experiment (DUNE)** from 2015 to 2023 and of **IceCube** as of 2023.

PUBLICATIONS

The following is a selected list of publications for which I was one of the primary contributors. Author lists are displayed alphabetically, as is the standard in particle physics. A complete list can be found at inspirehep.net/authors/1621061.

Citation metrics:

- **h-index** of 18 (publications with 10 authors or less) / 44 (all publications)
- **Total citations** 1,117 (publications with 10 authors or less) / 8,726 (all publications)

Top publications (with 10 authors or less):

1. Dark Neutrinos and a Three Portal Connection to the Standard Model, Peter Ballett, Matheus Hostert, Silvia Pascoli, Phys.Rev.D 101 (2020) 11 115025, arXiv:1903.07589 [hep-ph], citations: **105**.

2. MicroBooNE and the ν_e Interpretation of the MiniBooNE Low-Energy Excess, Carlos A. Argüelles, Ivan Esteban, Matheus Hostert, Kevin J. Kelly, Joachim Kopp, Pedro Machado, Ivan Martinez-Soler, Yuber F. Perez-Gonzalez, Phys.Rev.Lett. 128 (2022) 24 241802, arXiv:2111.10359 [hep-ph], citations: **100**.
3. Testing New Physics Explanations of the MiniBooNE Anomaly at Neutrino Scattering Experiments, Carlos A. Argüelles, Matheus Hostert, Yu-Dai Tsai, Phys.Rev.Lett. 123 (2019) 26 261801, arXiv:1812.08768 [hep-ph], citations: **87**.
4. Z' 's in neutrino scattering at DUNE, Peter Ballett, Matheus Hostert, Silvia Pascoli, Yuber F. Perez-Gonzalez, Zahra Tabrizi, Renata Zukanovich Funchal, Phys.Rev.D 100 (2019) 5 055012, arXiv:1902.08579 [hep-ph], citations: **87**.
5. Effective portals to heavy neutral leptons, Enrique Fernández-Martínez, Manuel González-López, Josu Hernández-García, Matheus Hostert, Jacobo López-Pavón, JHEP 09 (2023) 001, arXiv:2304.06772 [hep-ph], citations: **86**.

Pre-prints under review

1. Neutrino-Antineutrino Conversion from Ultralight Vector Dark Matter, Asher Berlin, Rodolfo Capdevilla, Ting Cheng, Matheus Hostert, Pedro A. N. Machado, 2026, submitted to arXiv.
2. New physics in multi-lepton tau decays, Yohei Ema, Patrick J. Fox, Matheus Hostert, Tony Menzo, Jure Zupan, 2026, arXiv:2604.01465 [hep-ph], under review by Phys.Rev.D.
3. Long-Lived Particles from Meson and Muon Decays at Rest at Spallation Sources, Matheus Hostert, Salvador Urrea, 2025, arXiv:2509.14085 [hep-ph], under review by Rev.Mod.Phys.
4. Resonant Neutrino Flavor Conversion in the Atmosphere, Connor Sponsler, Matheus Hostert, Ivan Martinez-Soler, Carlos A. Argüelles, 2024, arXiv:2405.12140 [hep-ph], citations: 2, under review by Phys.Rev.D.

Other peer-reviewed publications

6. Kaon decay constraints on vector bosons coupled to non-conserved currents, Matheus Hostert, Maxim Pospelov, Adrian Thompson, accepted for publication in Phys.Rev.D, arXiv:2602.19479 [hep-ph].
7. New Multi-messenger Probe of Dark Matter-Nucleon Interactions from Ultra-high Energy Cosmic Ray Acceleration, Stephan A. Meighen-Berger, P.S. Bhupal Dev, Matheus Hostert, accepted for publication in Phys.Rev.D, arXiv:2512.18093 [hep-ph].
8. Long-lived Axion-Like Particles from Tau Decays, Yohei Ema, Patrick J. Fox, Matheus Hostert, Tony Menzo, Jure Zupan, Phys.Rev.D 112 (2025) 11, 115028, arXiv:2507.15271 [hep-ph], citations: **13**.
9. Astrophysical flux of dark particles as a solution to the KM3NeT and IceCube tension over KM3-230213A, Yasaman Farzan, Matheus Hostert, JHEP 10 (2025) 208, arXiv:2505.22711 [hep-ph], citations: **15**.
10. Neutron portal to ultra-high-energy neutrinos, Gustavo F. S. Alves, Matheus Hostert, Maxim Pospelov, Phys.Rev.D 113 (2026) 7, 075035, arXiv:2503.14419 [hep-ph], citations: **16**.
11. Towards a Robust Exclusion of the Sterile-Neutrino Explanation of Short-Baseline Anomalies, Ohana Benevides Rodrigues, Matheus Hostert, Kevin J. Kelly, Bryce Littlejohn, Pedro A. N. Machado, Ibrahim Safa, Tao Zhou, Phys.Rev.Lett. 135 (2025) 8, 081801, arXiv:2503.13594 [hep-ph], citations: 2.
12. The Neutrino Slice at Muon Colliders, Luc Bojorquez-Lopez, Matheus Hostert, Carlos A. Argüelles, Zhen Liu, Phys.Rev.Lett. 135 (2025) 9, 091803, arXiv:2412.14115 [hep-ph], citations: 8.
13. Muon-induced baryon number violation, Patrick J. Fox, Matheus Hostert, Tony Menzo, Maxim Pospelov, Jure Zupan, Phys.Rev.D 110 (2024) 7 075015, arXiv:2407.03450 [hep-ph], citations: **11**.
14. Decaying sterile neutrinos at short baselines, Matheus Hostert, Kevin J. Kelly, Tao Zhou, Phys.Rev.D 110 (2024) 7 075002, arXiv:2406.04401 [hep-ph], citations: 3.

15. Panorama of new-physics explanations to the MiniBooNE excess, Asli M. Abdullahi, Jaime Hoefken Zink, Matheus Hostert, Daniele Massaro, Silvia Pascoli, *Phys.Rev.D* 111 (2025) 3 035028, arXiv:2308.02543 [hep-ph], citations: **19**.
16. New physics in multi-electron muon decays, Matheus Hostert, Tony Menzo, Maxim Pospelov, Jure Zupan, *JHEP* 10 (2023) 006, arXiv:2306.15631 [hep-ph], citations: **14**.
17. Pion decay constraints on exotic 17 MeV vector bosons, Matheus Hostert, Maxim Pospelov, *Phys.Rev.D* 108 (2023) 5 055011, arXiv:2306.15077 [hep-ph], citations: **23**.
18. Constraining light thermal inelastic dark matter with NA64, Martina Mongillo, Asli Abdullahi, Benjamin Banto Oberhauser, Paolo Crivelli, Matheus Hostert, Daniele Massaro, Laura Molina Bueno, Silvia Pascoli, *Eur.Phys.J.C* 83 (2023) 5 391, arXiv:2302.05414 [hep-ph], citations: **30**.
19. Semi-Visible Dark Photon Phenomenology at the GeV Scale, Asli M. Abdullahi, Matheus Hostert, Daniele Massaro, Silvia Pascoli, *Phys.Rev.D* 108 (2023) 1 015032, arXiv:2302.05410 [hep-ph], citations: **36**.
20. Implications of MicroBooNE's low sensitivity to electron antineutrino interactions in the search for the Mini-BooNE excess, Nicholas W. Kamp, Matheus Hostert, Carlos A. Argüelles, Janet M. Conrad, Michael H. Shaevitz, *Phys.Rev.D* 107 (2023) 9 092002, arXiv:2301.12573 [hep-ph], citations: **6**.
21. DarkNews: A Python-based event generator for heavy neutral lepton production in neutrino-nucleus scattering, Asli M. Abdullahi, Jaime Hoefken Zink, Matheus Hostert, Daniele Massaro, Silvia Pascoli, *Comput.Phys.Commun.* 297 (2024) 109075, arXiv:2207.04137 [hep-ph], citations: **11**.
22. Dipole-coupled heavy-neutral-lepton explanations of the MiniBooNE excess including constraints from MINERvA data, Nicholas W. Kamp, Matheus Hostert, Austin Schneider, Stefano Vergani, Melissa A. Uchida, *Phys.Rev.D* 107 (2023) 5 055009, arXiv:2206.07100 [hep-ph], citations: **31**.
23. Efficiently exploring multidimensional parameter spaces beyond the Standard Model, Carlos A. Argüelles, Nicolò Foppiani, Matheus Hostert, *Phys.Rev.D* 107 (2023) 3 035027, arXiv:2205.12273 [hep-ph], citations: **11**.
24. Dark sectors in neutron-shining-through-a-wall and nuclear-absorption signals, Matheus Hostert, David McKeen, Maxim Pospelov, Nirmal Raj, *Phys.Rev.D* 107 (2023) 7 075034, arXiv:2201.02603 [hep-ph], citations: **21**.
25. Heavy neutral leptons below the kaon mass at hodoscopic neutrino detectors, Carlos A. Argüelles, Nicolò Foppiani, Matheus Hostert, *Phys.Rev.D* 105 (2022) 9 095006, arXiv:2109.03831 [hep-ph], citations: **60**.
26. Novel multilepton signatures of dark sectors in light meson decays, Matheus Hostert, Maxim Pospelov, *Phys.Rev.D* 105 (2022) 1 015017, arXiv:2012.02142 [hep-ph], citations: **48**.
27. Constraints on decaying sterile neutrinos from solar antineutrinos, Matheus Hostert, Maxim Pospelov, *Phys.Rev.D* 104 (2021) 5 055031, arXiv:2008.11851 [hep-ph], citations: **26**.
28. A dark seesaw solution to low energy anomalies: MiniBooNE, the muon ($g - 2$), and BaBar, Asli Abdullahi, Matheus Hostert, Silvia Pascoli, *Phys.Lett.B* 820 (2021) 136531, arXiv:2007.11813 [hep-ph], citations: **78**.
29. Pair production of dark particles in meson decays, Matheus Hostert, Kunio Kaneta, Maxim Pospelov, *Phys.Rev.D* 102 (2020) 5 055016, arXiv:2005.07102 [hep-ph], citations: **36**.
30. Neutrino Masses from a Dark Neutrino Sector below the Electroweak Scale, Peter Ballett, Matheus Hostert, Silvia Pascoli, *Phys.Rev.D* 99 (2019) 9 091701, arXiv:1903.07590 [hep-ph], citations: **59**.
31. Neutrino trident production at near detectors, Matheus Hostert, *PoS NOW2018* (2019) 037, citations: **1**.
32. Neutrino Trident Scattering at Near Detectors, Peter Ballett, Matheus Hostert, Silvia Pascoli, Yuber F. Perez-Gonzalez, Renata Zukanovich Funchal, *JHEP* 01 (2019) 119, arXiv:1807.10973 [hep-ph], citations: **75**.

Community white papers and large collaborations

A list of community papers to which I contributed significantly, highlighting my participation below each entry.

1. First Search for Dark Sector e^+e^- Explanations of the MiniBooNE Anomaly at MicroBooNE, MicroBooNE collaboration, citations: **7**.
2. From oversimplified to overlooked: the case for exploring Rich Dark Sectors, Invited contribution to the Nuclear Physics B Special Issue on “Clarifying common misconceptions in high energy physics and cosmology”, 2025, arXiv:2505.05663 [hep-ph], citations: **2**.
3. Feebly Interacting Particles: FIPs 2022 workshop report, C. Antel et al, proceedings, 2023, arXiv:2305.01715 [hep-ph], citations: **222**.
4. The present and future status of heavy neutral leptons, Asli M. Abdullahi et al, J.Phys.G 50 (2023) 2 020501, arXiv:2203.08039 [hep-ph], citations: **324**.
5. Feebly-interacting particles: FIPs 2020 workshop report, Prateek Agrawal et al, Eur.Phys.J.C 81 (2021) 11 1015, 2021, arXiv:2102.12143, citations: **370**.
6. New opportunities at the next-generation neutrino experiments I: BSM neutrino physics and dark matter, C. A. Argüelles et al, Rept.Prog.Phys. 83 (2020) 12 124201, 2019, arXiv:1907.08311, citations: **124**.
7. Neutrino Non-Standard Interactions: A Status Report, P. S. Bhupal Dev et al, SciPost Phys.Proc. 2 (2019) 001, 2019, arXiv:1907.00991, citations: **259**.
8. Dark Sector Studies with Neutrino Beams, Brian Batell et al, proceedings, 2022, arXiv:2207.06898, citations: **41**.
9. The Physics Case for a Neutrino Factory, Alex Bogacz et al, proceedings, 2022, arXiv:2203.08094, citations: **15**.
10. White Paper on Light Sterile Neutrino Searches and Related Phenomenology, M. A. Acero et al, preprint, 2022, arXiv:2203.07323, citations: **141**.
11. Neutrino Self-Interactions: A White Paper, Jeffrey M. Berryman et al, proceedings, 2022, arXiv:2203.01955, citations: **133**.
12. New physics searches at kaon and hyperon factories, Evgueni Goudzovski, et al, Rept.Prog.Phys. 86 (2023) 1 016201, arXiv:2201.07805 [hep-ph], citations: **168**.
13. A Snowmass Whitepaper: Dark Matter Production at Intensity-Frontier Experiments, G. Krnjaic et al, preprint, 2022, arXiv:2207.00597, citations: **61**.

TALKS AND SEMINARS

Colloquia

- December 2025 — Columbia University, USA
- October 2025 — Idaho State University, USA
- January 2024 — University of Iowa, USA

Plenary Talks

- June 2026 (*upcoming*) — Neutrino 2026, USA
- May 2026 (*upcoming*) — New Physics Opportunities at Neutrino Experiments (NPN 2026), Ibaraki, Japan
- May 2026 (*upcoming*) — Phenomenology Symposium 2026, USA
- August 2025 — 2nd US Muon Collider Meeting, USA
- June 2025 — Interconnection between Particle Physics and Cosmology (PPC2025), USA

Invited talks

- April 2026 — APS Global Summit 2026, USA (Muon Collider session)
- April 2026 — APS Global Summit 2026, USA (Neutrino session)
- January 2026 — Neutrino Physics Center Mini-Workshop on Short-Baseline Neutrino Anomalies

January 2026	—	Neutrino Physics Center Seminar Series, Fermilab, USA
December 2025	—	Violation of Fundamental Symmetries with B mesons, Fermilab, USA
December 2025	—	Research Day at the Center for Neutrino Physics at Virginia Tech, USA
June 2025	—	NPN Cincinnati 2025, USA
June 2025	—	CIPANP 2025, USA
June 2025	—	Magnificent CEvNS 2025, Brazil (virtual)
May 2025	—	Mitchell Conference at Texas A&M, USA
March 2025	—	Aspen Winter Conference, USA
September 2024	—	Muon Collider Meeting by the Lake, USA
June 2024	—	Magnificent CEvNS 2024, Spain
May 2024	—	Mitchell Conference at Texas A&M, USA
April 2024	—	2nd Short-Baseline Experiment-Theory Workshop, USA
December 2023	—	Muons in Minneapolis workshop, USA
June 2023	—	PhenoBR 2023, Brazilian virtual conference
April 2023	—	American Physics Society April meeting, Minneapolis, USA
December 2022	—	NuTools workshop, Pittsburgh, USA
September 2022	—	ICTP Program on New Directions in Particle Physics, São Paulo, Brazil
June 2022	—	Neutrino Theory Workshop, NuTs, Madrid, Spain
March 2022	—	KITP, Interdisciplinary Developments in Neutrino Physics, Santa Barbara, USA
October 2021	—	Virginia Tech, neutrino seminar, Blacksburg, USA
September 2021	—	UK Muon Collider and NuSTORM meeting, UK
August 2021	—	vSTORM collaboration meeting, CERN
December 2020	—	Snowmass Dark Sector Studies at High Intensities Frontier, USA
November 2020	—	Central American meeting of High Energy Physics, Cosmology and High Energy Astrophysics, Cidade da Guatemala, Central America
October 2020	—	3rd South American Dark Matter Workshop, ICTP, São Paulo, Brazil
October 2020	—	PIKIMO 9, Kentucky, Kentucky, USA
October 2020	—	Snowmass Baryon and Lepton Number Violating Processes workshop, USA
September 2020	—	Snowmass Theory of neutrino physics mini-workshop, USA
September 2020	—	Snowmass Neutrino Frontier 03 kick-off meeting, USA
December 2019	—	NuPhys 2019, London, UK
October 2019	—	CERN Neutrino Platform Week 2019, CERN, Switzerland
May 2019	—	Neutrino Theory Network Workshop, Washington U., St Louis, St Louis, USA
April 2019	—	Prospects of Neutrino Physics, IPMU, Kashiwa, Japan
December 2018	—	Physics Opportunities at the Near Detector of DUNE (PONDD), Fermilab, Fermilab, USA
June 2018	—	Near detector workshop 2018, CERN, CERN, Switzerland

Invited parallel talks

November 2021	—	Brookhaven Forum 2021, Brookhaven National Laboratory, USA
September 2018	—	Neutrino Oscillation Workshop 2018, Ostuni, Italy
August 2018	—	NuFact 2018, Virginia, Blacksburg, USA
May 2018	—	Phenomenology Symposium 2018, Pittsburgh, USA

Parallel and contributed talks

February 2025	—	CERN Neutrino Platform Pheno Week, CERN, Switzerland
August 2024	—	Inaugural US Muon Collider Community Meeting, Fermilab, USA
June 2024	—	Magnificent CEvNS
October 2022	—	Feebly Interacting Particles Workshop 2022, CERN, Switzerland
September 2022	—	CIPANP 2022, Orlando, USA
August 2022	—	TeVPA 2022, Kingston, Canada

July 2022 — Snowmass 2022, Seattle, USA
 July 2021 — American Physics Society Division of Particles and Fields meeting, USA
 April 2021 — American Physics Society April Meeting, USA
 February 2021 — XIX International Workshop on Neutrino Telescopes, Italy
 July 2020 — ICHEP 2020, Prague, Czech Republic
 June 2020 — Neutrino 2020, University of Chicago, Chicago, USA
 May 2020 — Phenomenology Symposium 2020, Pittsburgh, USA
 June 2019 — Invisibles Workshop 2019, Valencia, Spain

Invited seminars

December 2025 — University of Wisconsin, USA
 November 2025 — Grinnell College, USA
 October 2025 — Sydney Consortium for Particle Physics and Cosmology, Australia
 October 2025 — Brookhaven National Lab, USA
 March 2025 — Argonne National Laboratory, USA
 October 2024 — University of Iowa, USA
 September 2024 — Northwestern University, USA
 September 2024 — Fermilab Theory Division, USA
 October 2023 — University of Massachusetts, Amherst, USA
 April 2023 — Columbia University, New York, USA
 April 2023 — California Institute of Technology, Pasadena, USA
 December 2022 — Northwestern University, Evanston, USA
 December 2022 — Los Alamos National Laboratory, Los Alamos, USA
 November 2022 — Queen's University, Kingston, Canada
 November 2022 — King's College London, London, UK
 November 2022 — Carleton University, Carleton, Canada
 March 2022 — University of Texas A&M, College Station, USA
 March 2022 — University of Toronto, Toronto, Canada
 February 2022 — TRIUMF/University of Victoria, Victoria, Canada
 January 2022 — University of Kentucky, Kentucky, USA
 December 2021 — SLAC, Stanford National Laboratory, USA
 November 2021 — Harvard University, family meeting, Boston, USA
 September 2021 — Perimeter Institute, particle physics seminar, Waterloo, Canada
 May 2021 — McGill University, Montreal, Canada
 April 2021 — ETH, Zürich, Switzerland
 April 2021 — C3P, UCLouvain, Louvain, Belgium
 April 2021 — Carleton University, Carleton, Canada
 March 2021 — University of California Santa Cruz, Santa Cruz, USA
 March 2021 — Neutrino Seminar, Fermilab, Fermilab, USA
 June 2020 — JGU Theorie Palaver, Mainz, Germany
 May 2020 — Brookhaven Neutrino Theory Virtual Seminars, Brookhaven National Laboratory, USA
 February 2020 — Fermilab Theory Seminar, Fermilab, USA
 August 2019 — Columbia University, New York, USA
 August 2019 — MicroBooNE collaboration call, USA
 May 2019 — IFIC, Valencia, Spain
 March 2019 — Queen Mary University of London, London, UK
 November 2018 — Max-Planck-Institut für Kernphysik, Heidelberg, Heidelberg, Germany
 June 2018 — Perimeter Institute, Waterloo, Canada
 May 2018 — Fermilab Theory Seminar, Fermilab, USA